Streamlining Success through "Intelligent" Enterprise Project Management Platform

Technical Architecture Document Draft

March 2026

Requirements Areas

Innovation Management	Idea & Demand Management	Project Management	Time & Resource Management	Visibility & Dashboarding	Vendor Management
Intrapreneurship	Idea /Demand Submission	Project Initiation	Resource Planning	Integrated Project Dashboards	Vendor Evaluation
Start-up/POC Tracking		Project Planning	Time Sheet Management		Vendor Performance Tracking
		Change Management			
		Risk Management			
		Project Closure			
		Benefit Tracking			

Current Maturity Level

Level	Process	System
I	✗	✗
II	✓	✗
III	✓	✓

IBM | MARUTI SUZUKI

Functional Requirements

Please Refer to The Functional Requirements Deck

Non-Functional Requirements

Category	Non-functional requirement	Comments	Specifications			
System Load			**Current**	**Future in 1 year**	**In 2 years**	**In 3 years**
	Total number of users of the system		30,000			
	Number of concurrent users at peak time		3,000			
	Number of concurrent users (average)		1,800			
Response Time	Target & Maximum Response Time (seconds) for different categories (add categories as per project need)					
	Landing page / Main Screen		<500 ms			
	Normal pages		<500 ms			
	Dashboard/heavy processing page		1 sec			
	Reports		1 sec			
	Threshold for service API (for non UI applications)		<500 ms			
	Target acceptance criteria like response time on 95 percentile of requests		6 sec			
Performance & Integration	Availability & Reliability					
	Business-as-usual uptime SLA (e.g., 99.9% / 99.95%)		99.90%	99.90%	99.90%	99.90%
	Planned vs unplanned downtime	Planned- max 1 hour/month unplanned- <=0.1% monthly downtime				
	Active-active / active-passive setup					
	Scalability					
	Horizontal vs vertical scaling					
	Auto-scaling support					
	Observability & Monitoring					
	Application metrics (CPU, memory, latency)	Server CPU, Memory, Latency.				
	Centralized logging- Operational logs	Yes				
	Alerting & escalation	Yes				
	Dashboard availability (Grafana/App Insights/etc.)	App Dynamics , API monitoring				
	Integration & Interoperability					
	API versioning	Required				
	External dependency SLAs	<1.5 sec for SaaS based API response time				
	Retry & timeout policies	3 retries and				
	DevOps & Release Management					
	Zero-downtime deployment	yes				
	Rollback strategy	Yes				

Non-Functional Requirements

Category	Non-functional requirement	Comments	Specifications
Privacy & Data Protection			
	PII identification		
	Consent management		
	Data residency requirements		
User Experience	Browser supported with version		
	Chrome (latest + last 2 versions)	Yes	
	Microsoft Edge (Chromium, latest + last 2)	Yes	
	Safari (macOS & iOS – last 2 versions)	Yes	
	Look and feel		
	Desktop	Yes	
	Tablet	Yes	
	Mobile	Yes	
	Orientation support (portrait/landscape)	Yes	
	Internationalization and localization (application level / code level / log level)		
	Languages supported	All	
	No. of emails/SMS to be translated per second		
	Date time display	Yes	
	Currency	All	
	Time-zone handling	All	
	Accessibility requirements		
	Text to speech	Yes	
	Speech to text	Yes	
	WCAG 2.1 AA compliance	Yes	
Security			
	Database encryption needed specify the tables, fields etc.	Encrypted at Rest and Transit	
	Encryption mechanism to be used in Application	Specify	
	Communication between App, DB or other internal server needs to be controlled?		
	Communication with external systems needs to be controlled?		
	Communication protocol for external system integration	REST	
	Authentication mechanism for external system integration	OAuth2	
	SSL required for external APIs?	Yes	
	User is not allowed to access any url directly without login	Yes	
	Security threats like SQL injection	Yes	
	Security acceptance criteria like HCL AppScan		
	Role-based access control (RBAC)	Yes	
	Maker–checker approval workflows	Yes	

Non-Functional Requirements

Category	Non-functional requirement	Comments	Specifications			
Audit and Edit trail						
	Audited elements- What business elements will be audited?	User master, role & access configurations, Customer id, transactions, approval workflow, configuration changes, reference data, policy/rule configuration				
	Audited fields- What data fields will be audited?	Action type (create/update/delete etc), Fields name, old/new value, timestamp, roles, approval status, access, PII fields etc				
	Audit file characteristics (Audit & compliance logs)- Timestamp, user id, role	User ID, Timestamp				
	Approval logs (who approved what, when)	Yes				
	Elements- What elements history needs to be maintained	Version history, records to be retrieved, system must support comparison between versions				
	Audit Monitoring & Compliance Reporting	Yes				
Recovery and Backup						
	Custom Recovery process needed?	Yes				
	Disaster scenarios recovery timescales- How quickly does recovery takes to perform/ down time acceptable	RPO and RTO need to be specified in hours. RPO refers to the maximum duration of data loss that can be tolerated. RTO refers to the time required to resume operations after a disaster, typically from the Disaster Recovery (DR) site.				
	Backup frequency for disaster recovery - How often is backup taken for	Once a day or twice a day.				
	- Database (like Weekly full and Daily/ hourly transactional)	Once a day or twice a day.				
	- Folders and files	Once a day or twice a day.				
	- Set-up data	Once a day or twice a day.				
	- System state	Once a day or twice a day.				
	- Code	Once a day or twice a day.				
	Database purging / archival policy	Years or months.				

Non-Functional Requirements

Category	Non-functional requirement	Comments	Specifications			
Regulatory Compliance			**Current**	**Future in 1 year**	**In 2 years**	**In 3 years**
	Financial regulations (based on geography): SEBI (India), SEC / FINRA (US), GDPR / DPDP Act?	yes				
	Data retention & archival rules	yes				
	Audit log retention period	Years or months.				
Document Management						
	Maximum number of documents per day		50	55	60	70
	Average size of each document		15-20 MB	20-25 MB	25-30 MB	30-50 MB

Users in the System

1. Demand Manager	2. DE Business Partner	3. Program Manager	4. Project Manager	5. COE Head
6. Resource Manager	7. Risk Controller	8. Governance Admin	9. Vendor Manager	10. QA Manager
11. Release Manager	12. KPI Admin	13. Sprint Manager	14. WBS Admin	15. JIRA Admin
16. Timesheet Admin	17. Finance	18. Legal	19. Business User	20. Team Member
21. Timesheet User	22. Ecosystem Partner	23. Startup	24.	25.

Critical Controllers
High complexity and + cross-module control + high governance impact

Operational Orchestrators
High activity volume + multi-touchpoint execution ownership, require dashboards

Specialized Owners
Deep module Expertise with focused scope

Transactional Users
Low complexity, high volume, Enter Data, Trigger Workflows, Limited Decision Authority

External Actors
Participate in selective stages Limited system ownership

Architecture Considerations

Scalability & Performance

- System should support horizontal and vertical scaling.
- Stateless service design where possible.
- Caching strategy (Redis/Elasticache) for high-traffic modules.

High Availability & Resilience

- Graceful handling of service failures.
- Circuit breaker pattern for downstream dependencies.

Security & Compliance

- Role-based and attribute-based access control.
- JWT/OAuth 2.0 authentication.
- API rate limiting, throttling, input sanitization.
- Audit logs for all CRUD operations.
- Data encryption (in transit + at rest).

Business Services

- System to follow **modular microservices principles**, including:

Service Decomposition

- Ideally separate services by domain/function:
- Project Service
- Task Management Service
- Resource Allocation Service
- Reporting & Analytics Service
- Notification Service
- User & Role Management Service
- Integration Service (SAP/ERP/AD/Other MSIL systems)

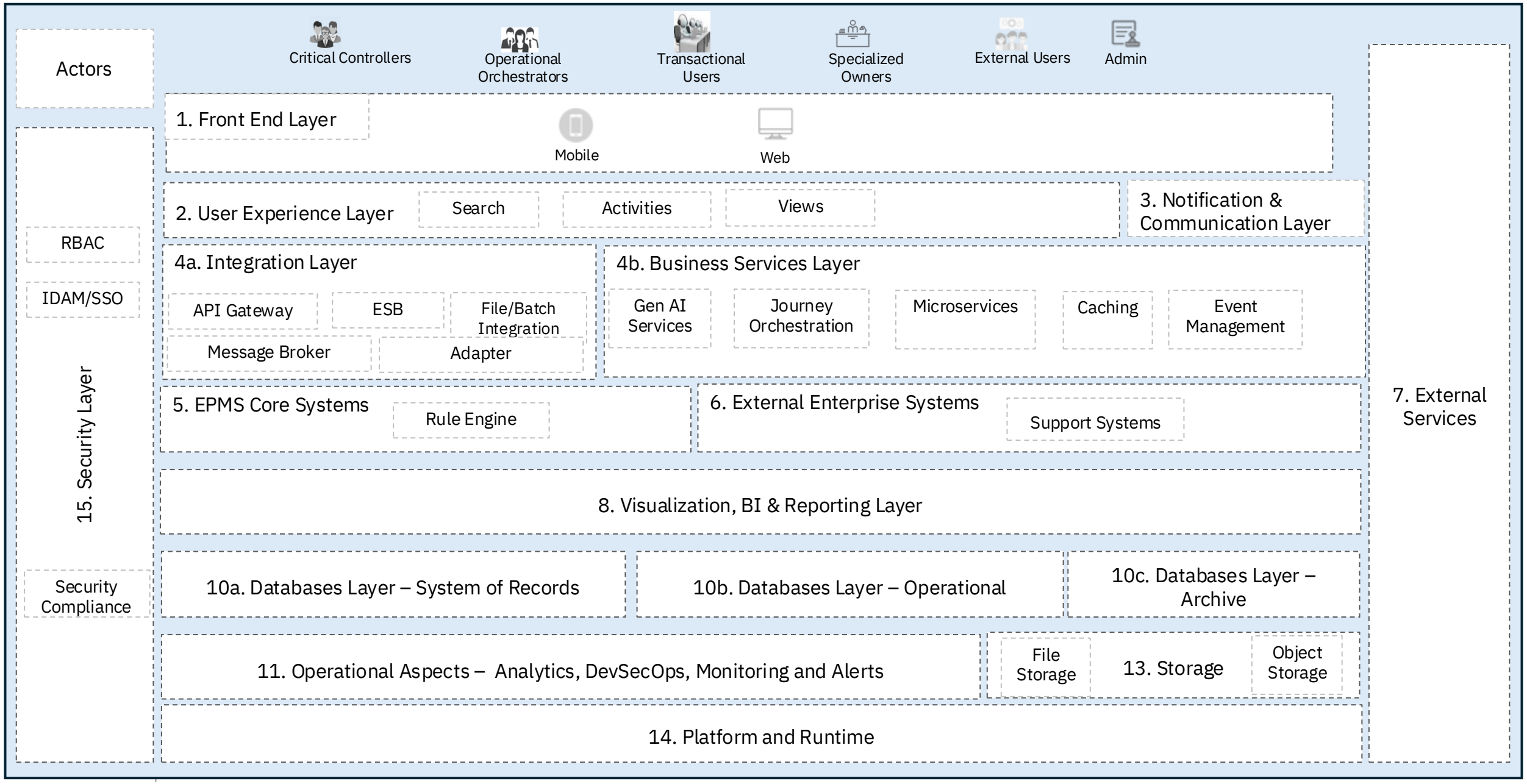
Technology & Framework Considerations

- Container-based deployment (Docker + Kubernetes).
- CI/CD pipelines
- Logging & monitoring using ELK, Prometheus, Grafana.
- Prefer open standards to simplify future integrations.

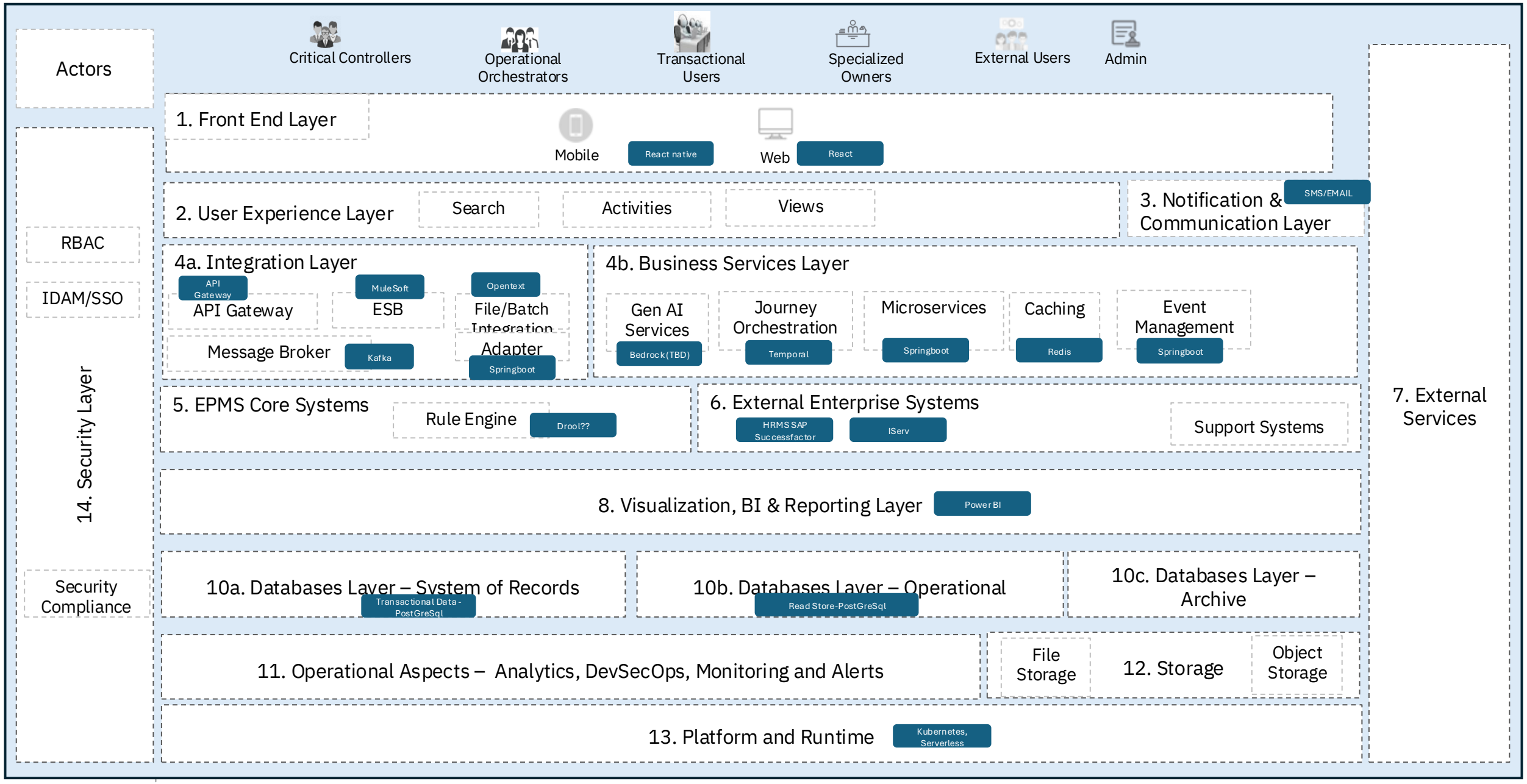
Integration Requirements

- Architecture must support easy integration with:
- Existing enterprise systems (Common Services, AD, etc.)
- Notification channels (Email, SMS, WhatsApp)
- API-based data exchange with other internal tools

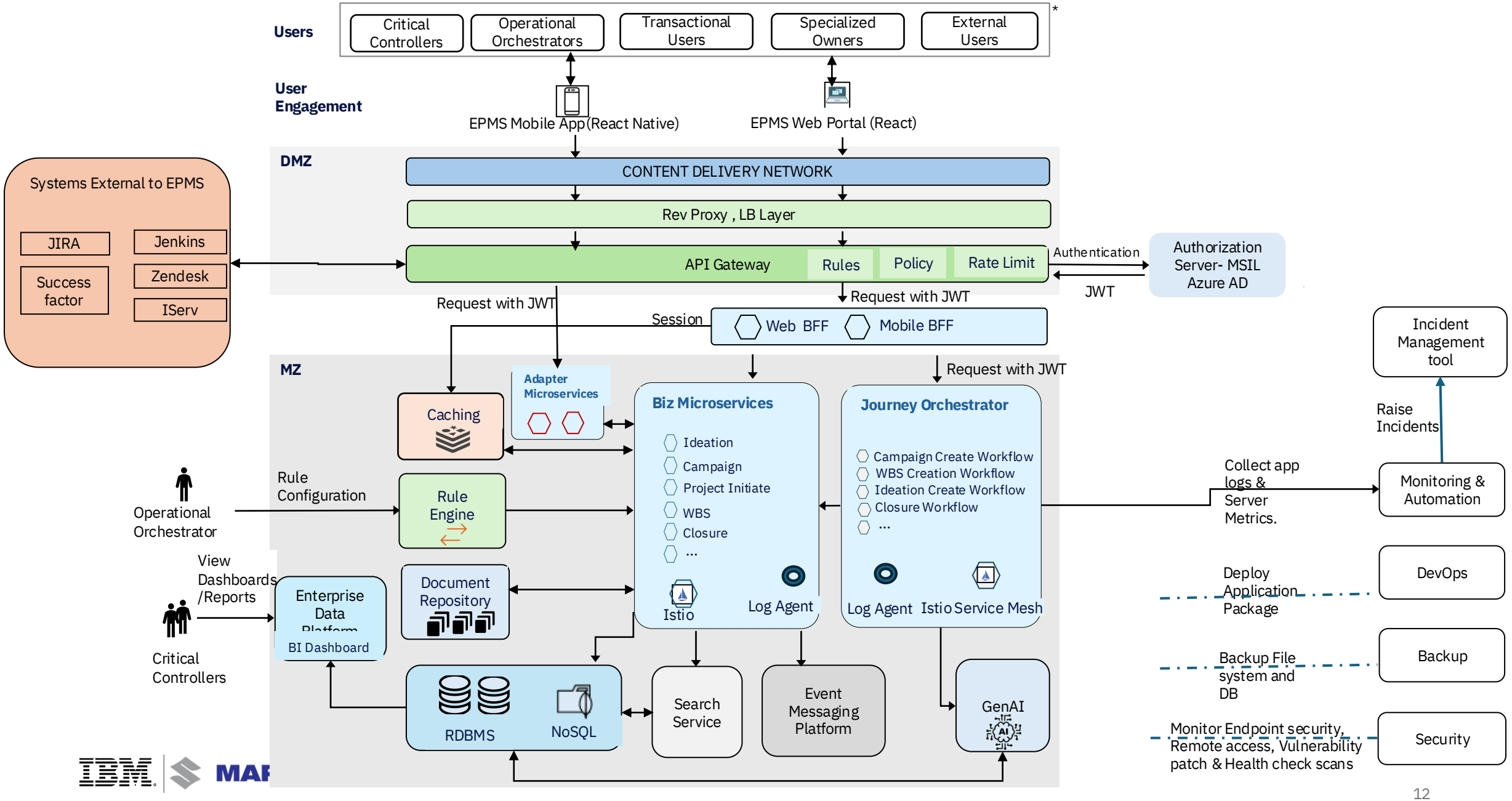
EPMS Architecture Building Blocks



EPMS Architecture Building Blocks Overlay



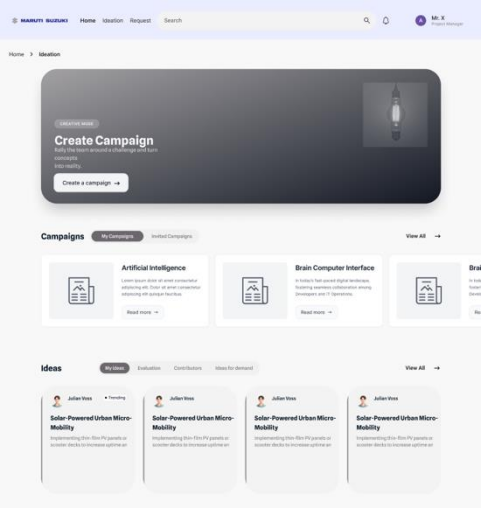
EPMS Architecture

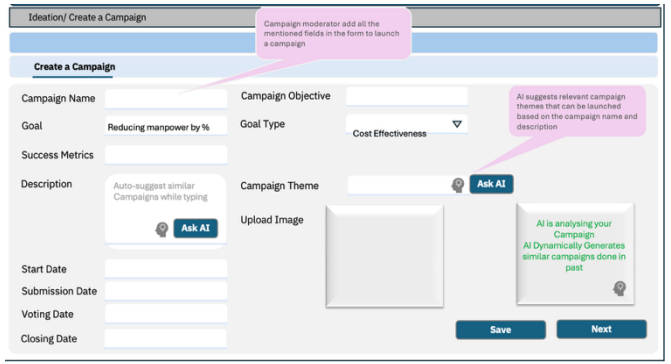


*Indicative Personas reference are Mentioned in the later slide

Campaign Management Design

Campaign Management





Users

- CoE Head
- DE Business Partner

Campaign Management Features

- Create New Campaign
- Campaign listing with “My campaigns”, invited campaigns, “View-all” with filters (Date).
- Search Campaigns
- Detail of campaigns based on selected campaign from the list.
- Suggest themes and similar past campaigns while typing.
- Notifications - Invite, phase changes, reminders.
- Audit & history - Who changed what and when.

Campaign - Journeys

Features

- Launch new campaign/challenge
- Suggest themes and similar past campaigns while typing.
- Publish Campaign for ideation.

- Campaign listing with “My campaigns”, invited campaigns, “View-all” with filters (Date).
- Search Campaign
- Detail of campaigns based on selected campaign from the list.

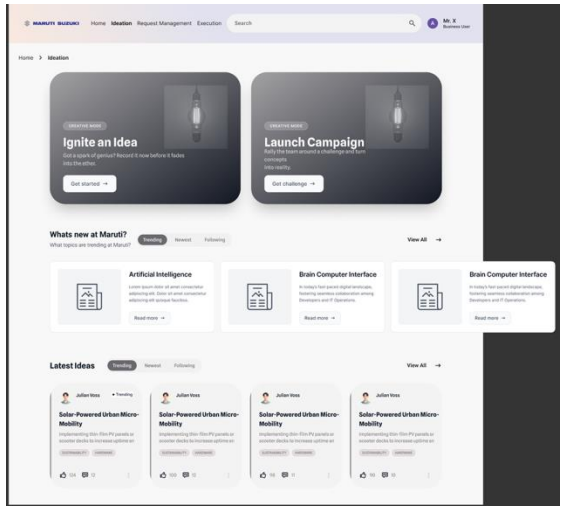
Journeys

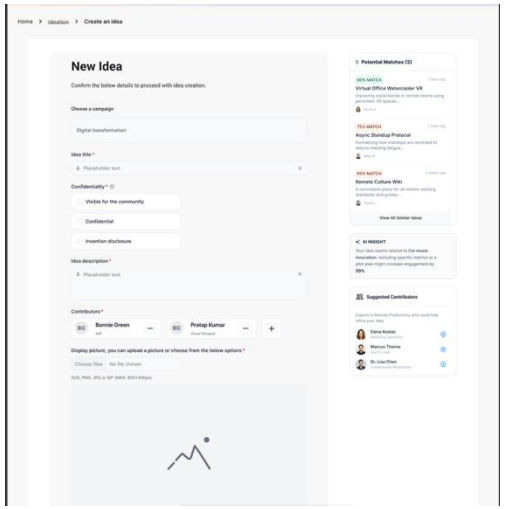
Launch Campaign Journey

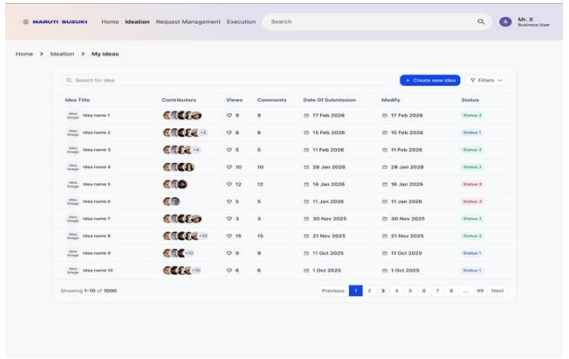
Search Campaign Journey

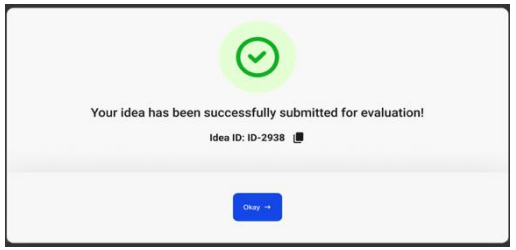
Ideation Design

Ideation - Features









Users

- User
- Campaign Moderator
- Evaluator

Idea Creation Features

- Idea creation against a campaign
- Idea submission success with generated Idea ID
- Rating for an Idea
- Action for an idea
- Idea listing with Status , Contributors, Views , Comments, Submission Dates
- Search Ideas from the listing
- Potential Match with the Older Ideas
- View Similar Ideas
- View Latest Ideas
- AI-based similarity matching for Ideas

Ideation - Journeys

Features

- Create idea
- Success with generated Idea ID.
- Rate and Idea
- Action Taken for an Idea
- Send notifications - Invite, phase changes, reminders.
- Capture Audit & history - Who changed what and when.

- List ideas
- View similar ideas
- AI similarity for Ideas entered
- Potential match with older ideas

Journeys

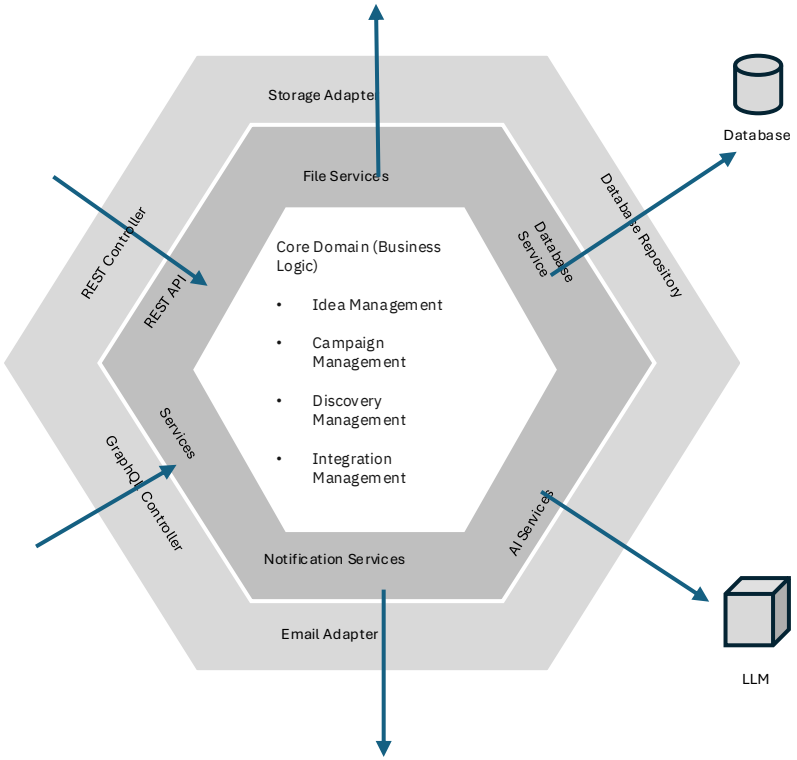
Create Idea Journey

Search Ideas Journey

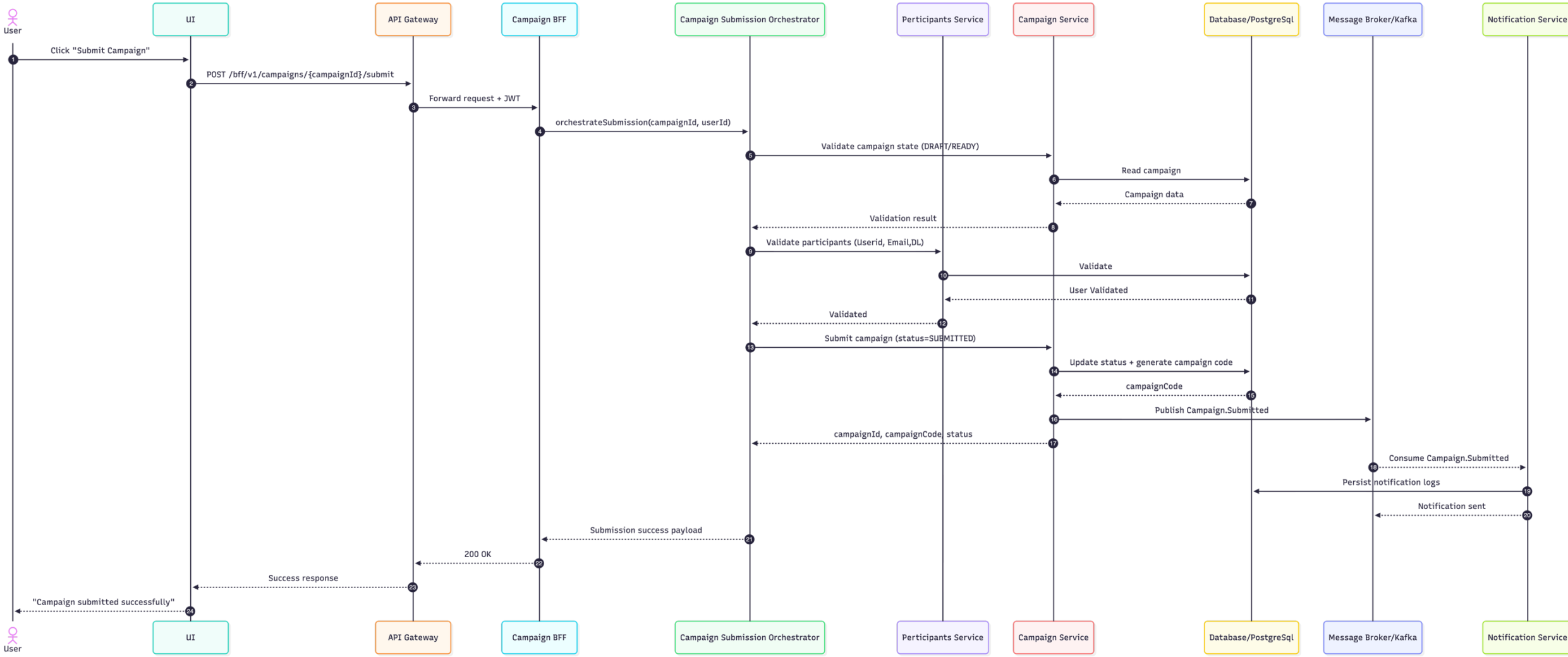
Campaign And Ideation Management– Domains and Microservices

Domain	Scope	Key Capabilities
Idea Domain	Handles idea creation and submission lifecycle	Create idea, set confidentiality, Create idea, generate Idea ID, Listing ideas
Campaign Domain	Handles campaign launch and publication lifecycle	Create campaign, publish campaign, list campaigns , Generate Campaign ID
Integration Domain (Common)	Handles external adapters and integrations	Notifications Services

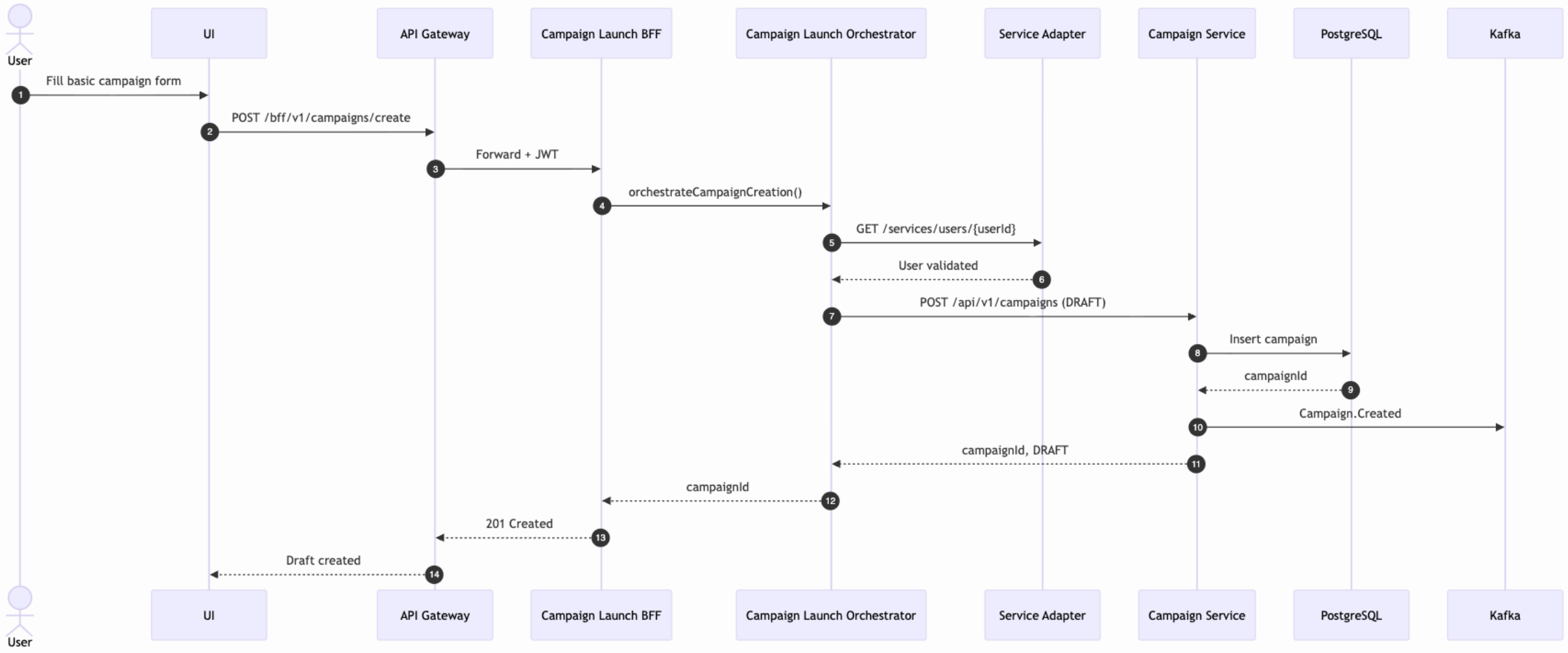
Microservice	Type	Domain	Key Responsibility
Idea Service	Business	Idea	Idea CRUD, status management, All read queries
Campaign Service	Business	Campaign	Campaign CRUD, All read queries
Service Adapter	Adapter	Integration	Fetch user profiles from HRMS, Storage, External AI services, Email/SMS
Campaign BFF	BFF	Cross-domain	Aggregate Idea and campaign journey
Idea Submission Orchestrator	Orchestrator	Cross-domain	Coordinate idea submission workflow
Campaign Launch Orchestrator	Orchestrator	Cross-domain	Coordinate campaign launch workflow



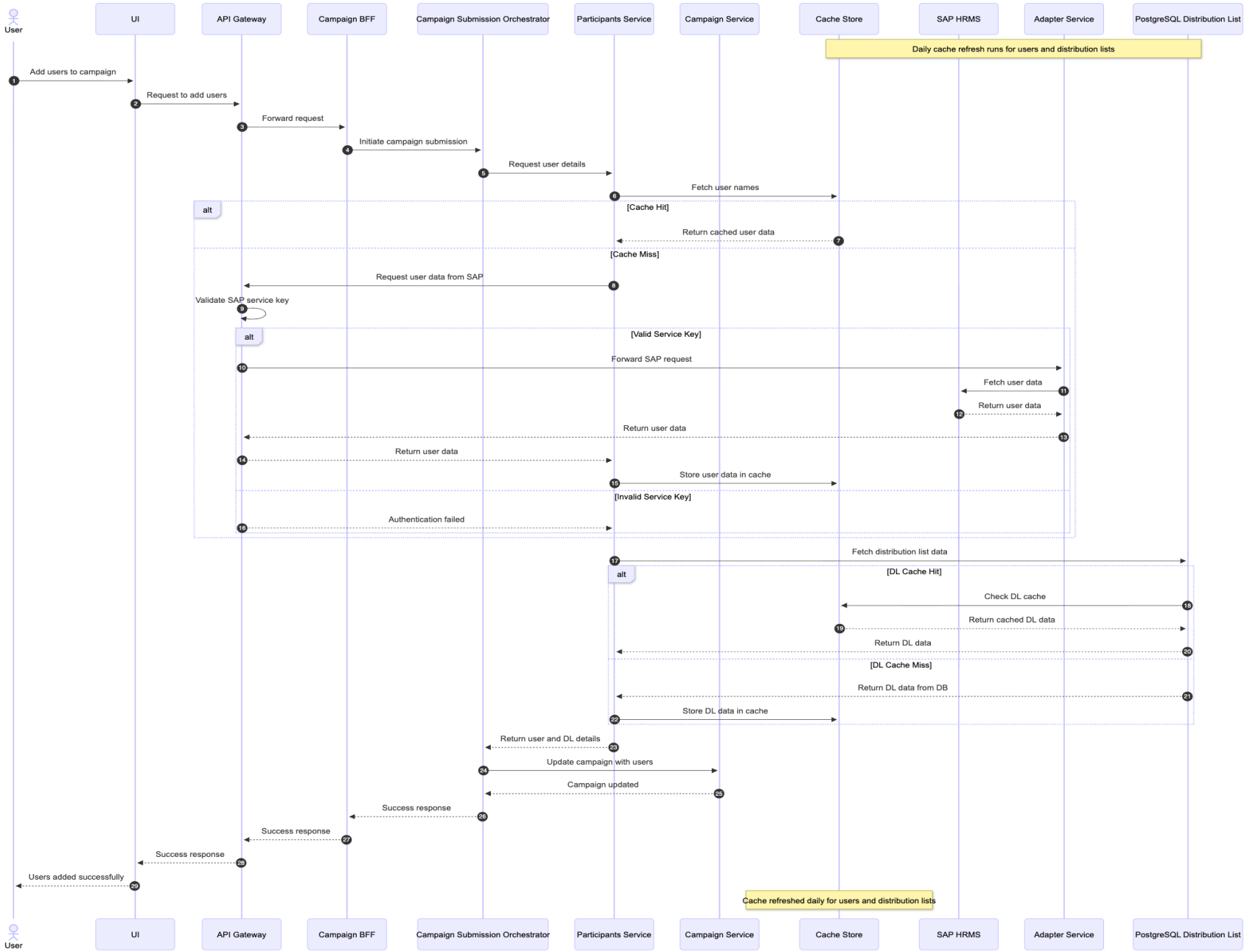
Sequence Diagram Journey: Launch Campaign – Overall Journey



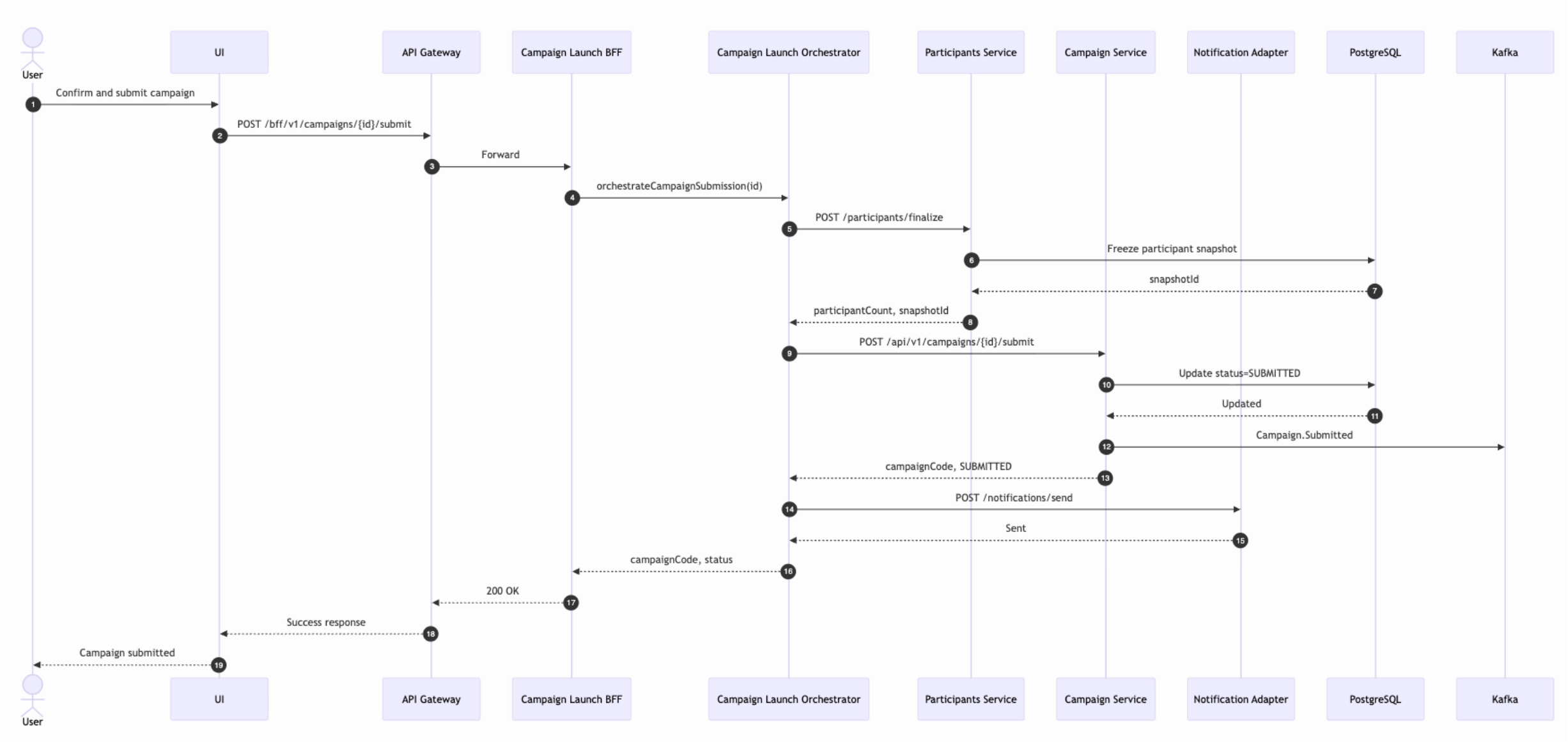
Sequence Diagram Journey: Launch Campaign – Campaign Creation



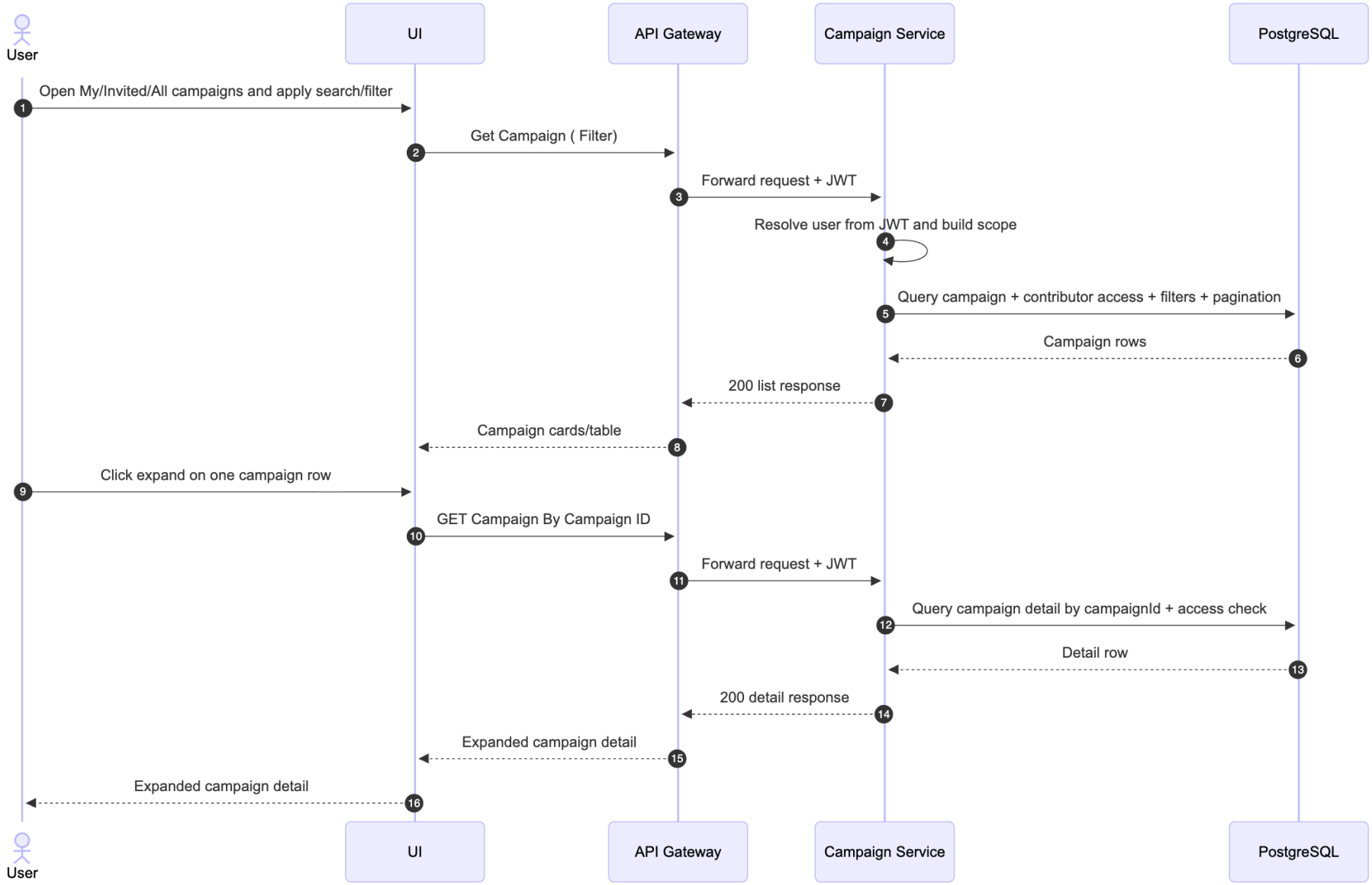
Sequence Diagram : Journey: Launch Campaign – Search User ,DL for Campaign



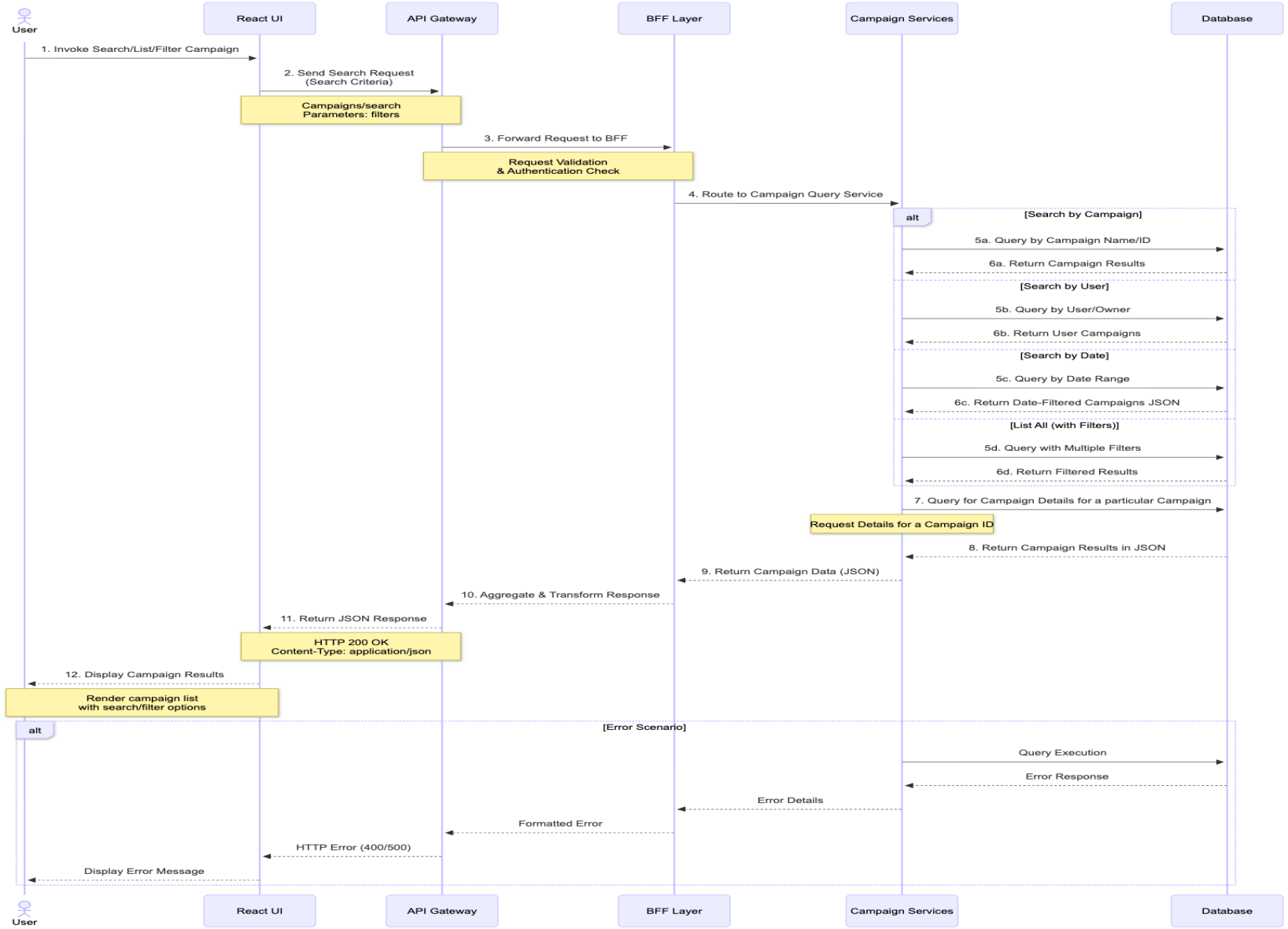
Sequence Diagram : Journey: Launch Campaign – Save Campaign



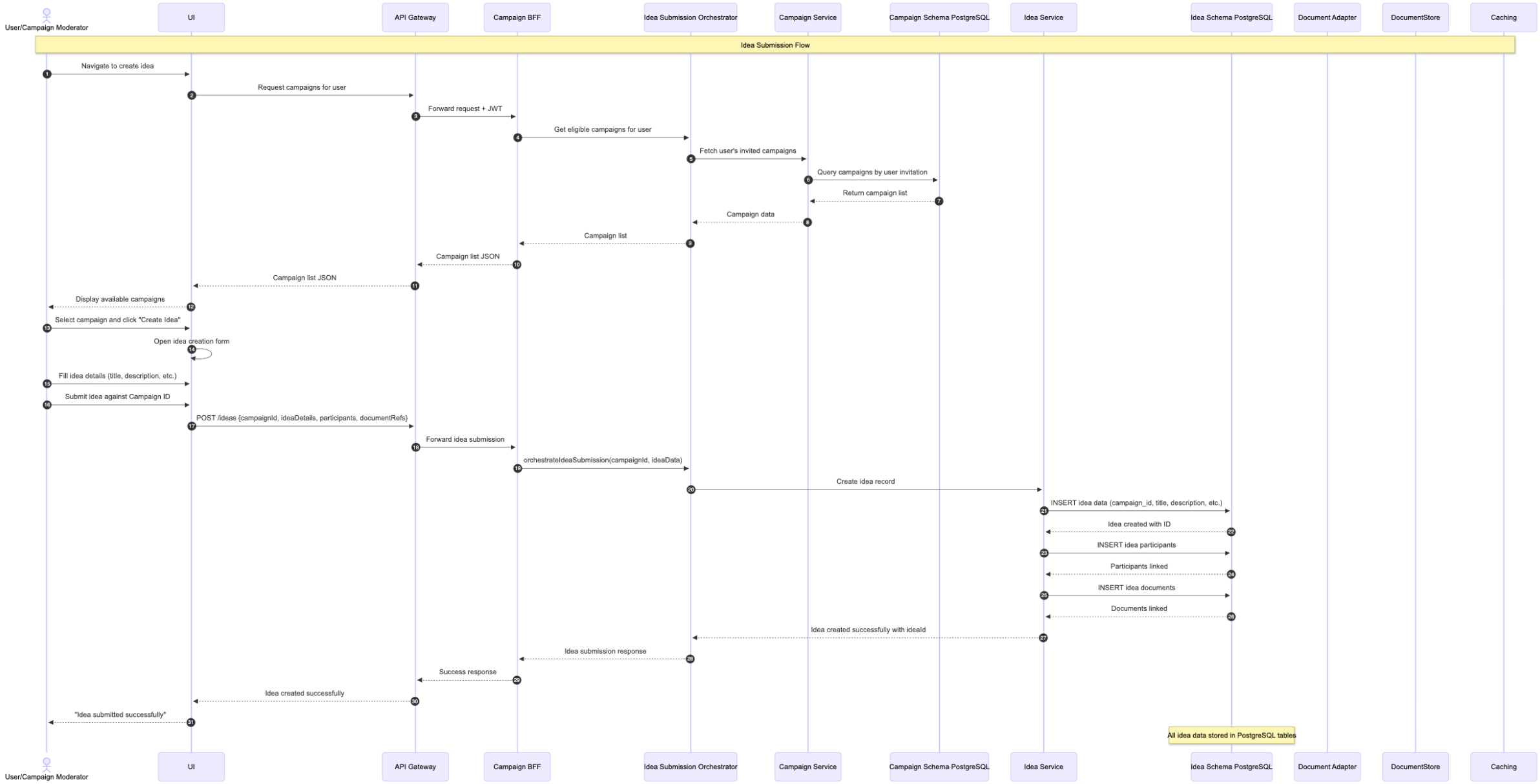
Sequence Diagram : Journey: View/Search/Filter My Campaign/Invited/All Campaigns



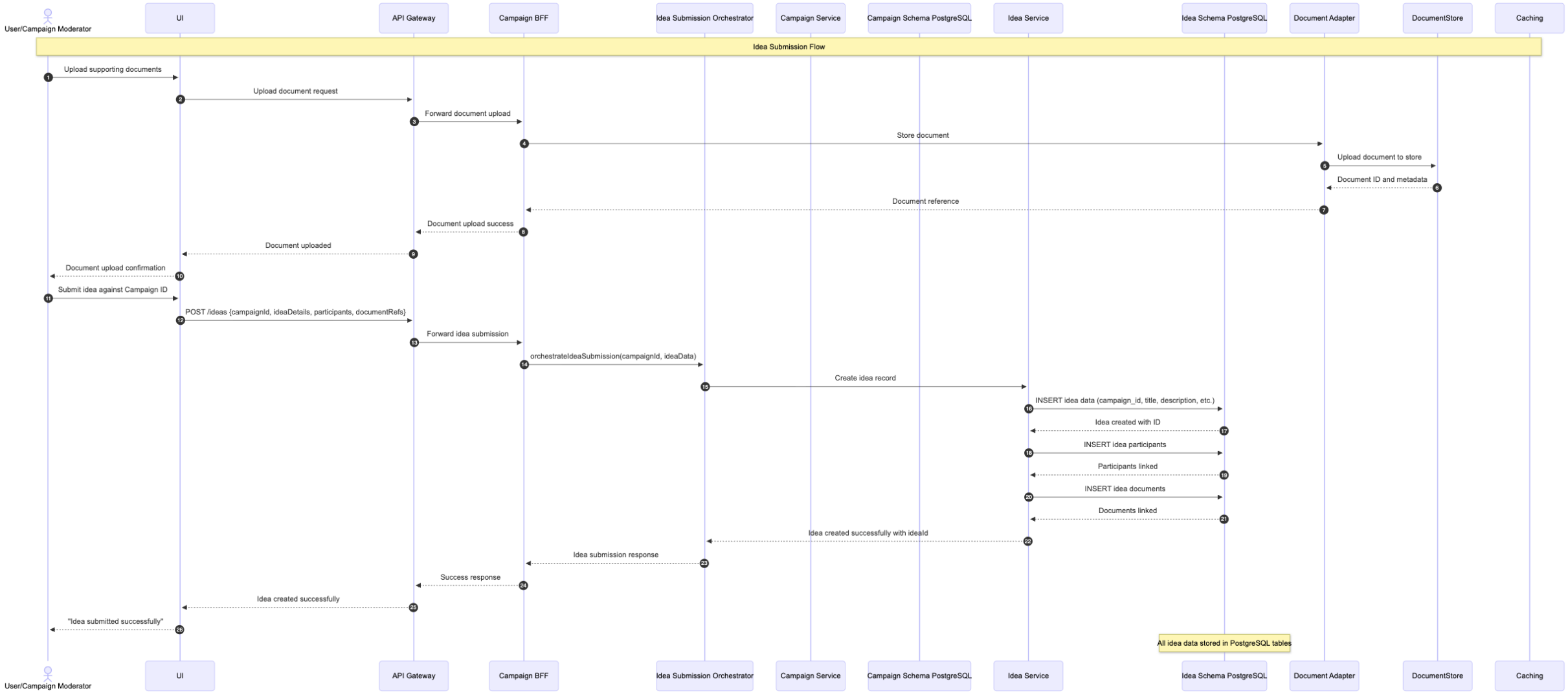
Sequence Diagram : Journey: Expand/View Campaign Detail



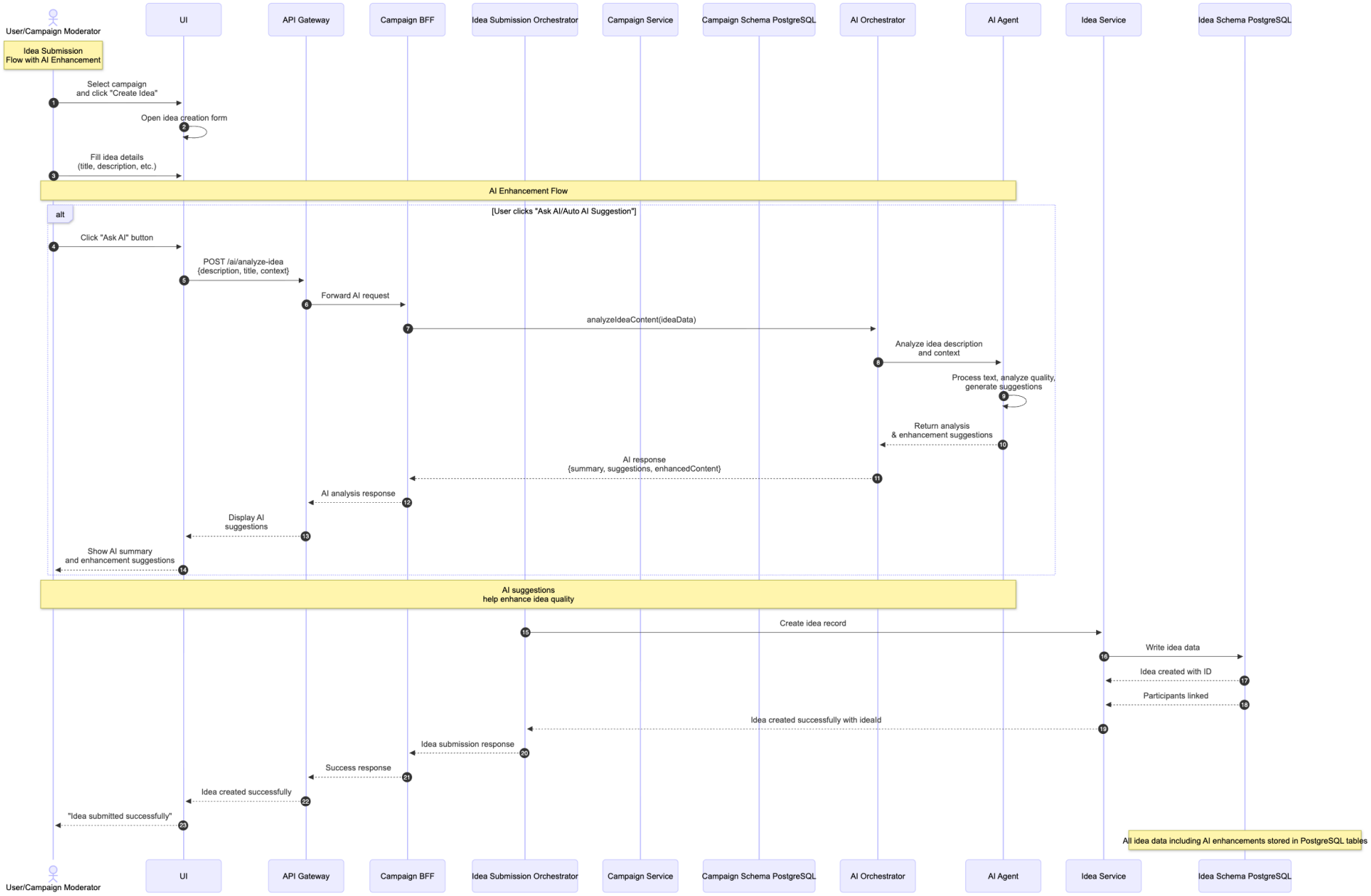
Sequence Diagram Journey: Idea Creation



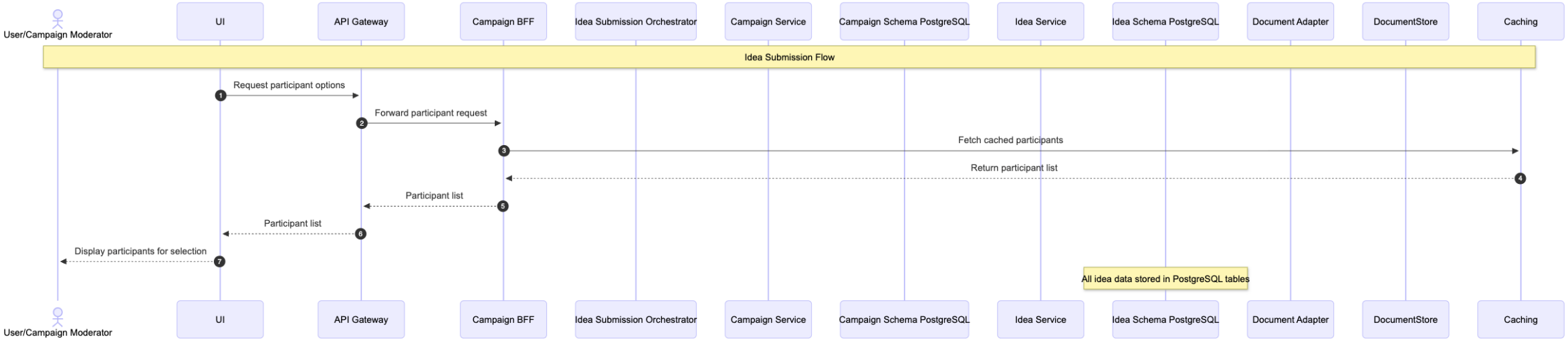
Sequence Diagram Journey: Idea Creation (Document Upload Sub Flow)



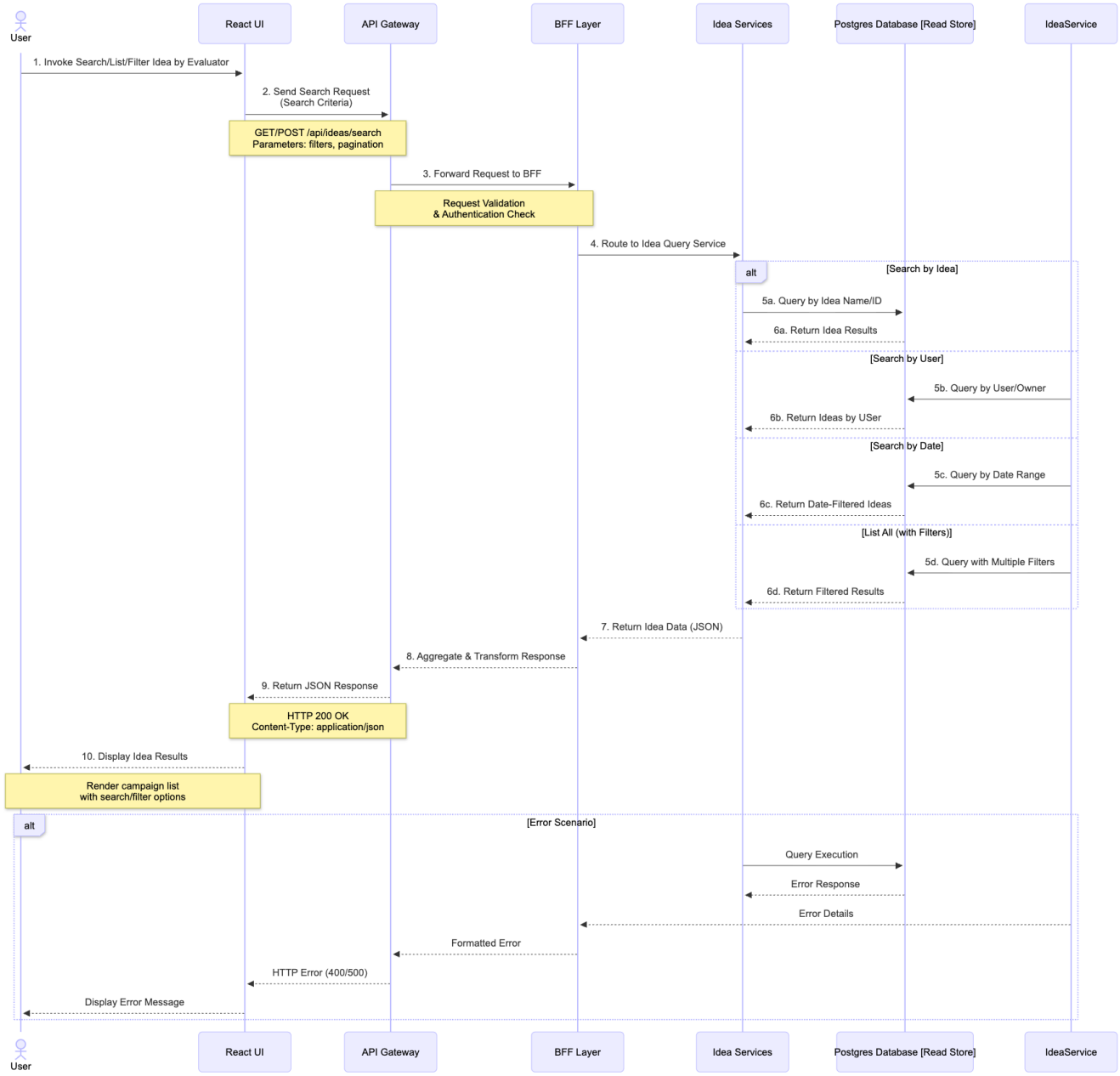
Sequence Diagram Journey: Idea Creation (Gen AI - Sub Flow)



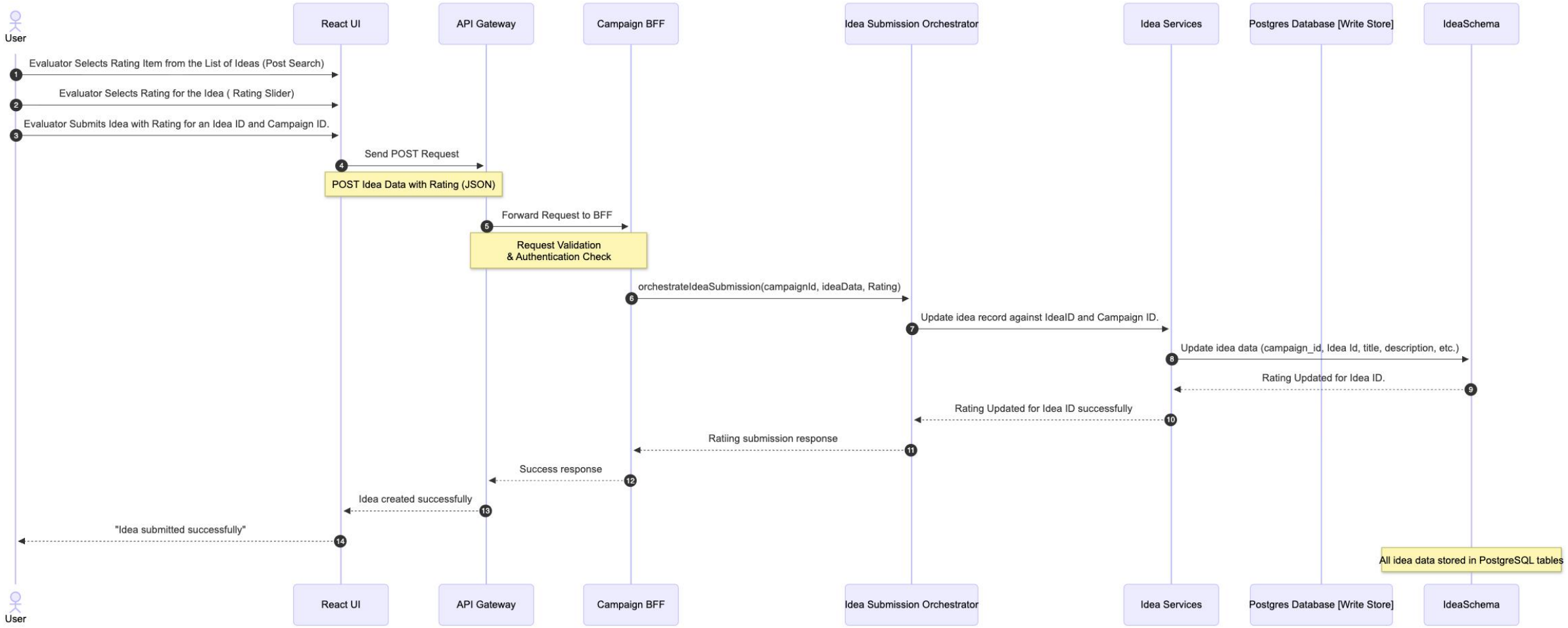
Sequence Diagram Journey: Idea Creation (User Add/edit Sub flow)



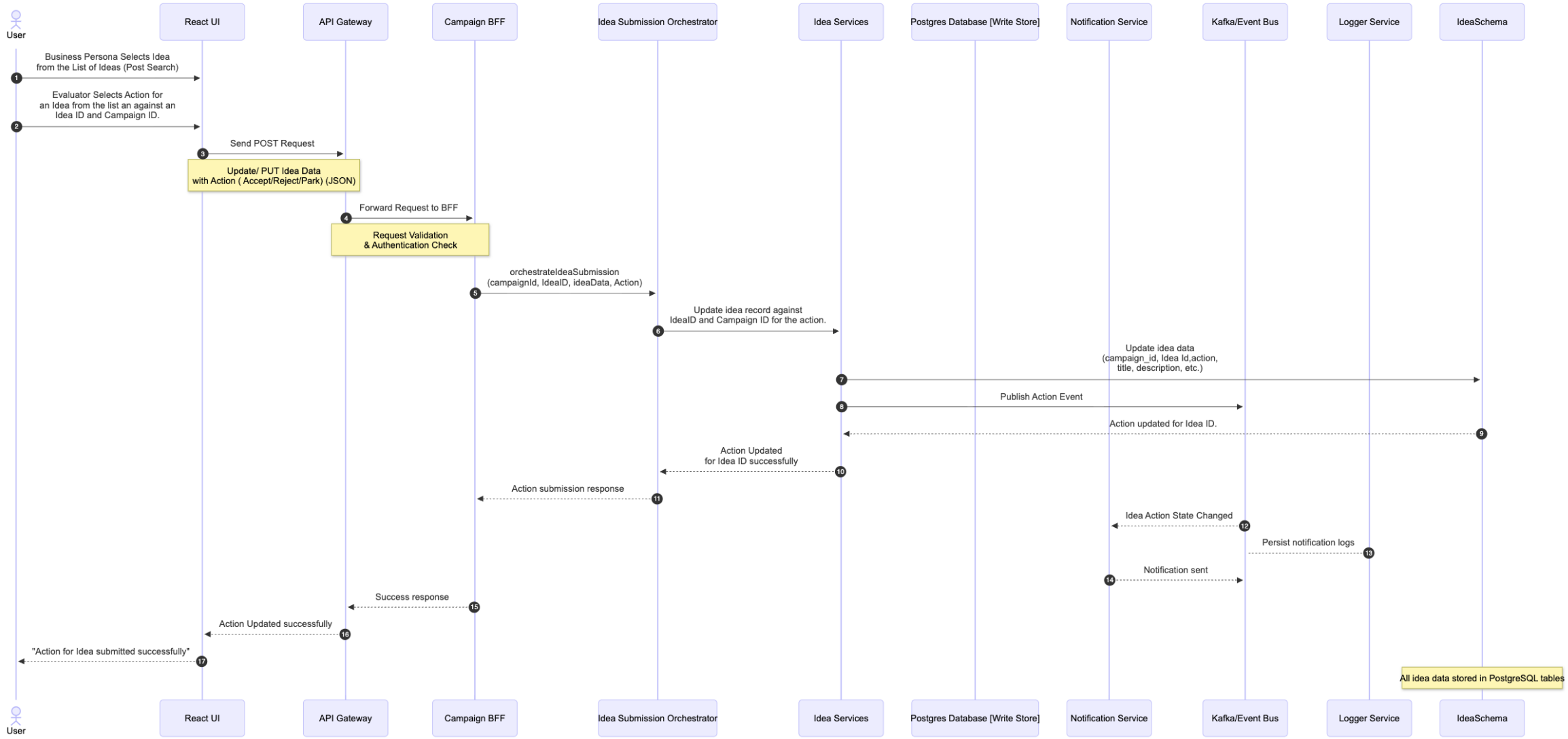
Sequence Diagram Journey: Search Ideas



Sequence Diagram Journey: Rating of Ideas by the Evaluator



Sequence Diagram Journey: Action Taken for Ideas by the Business Person



Campaign – Data Tables

campaigns

Attribute Name	Data Type	Constraints	Journey Mapping	Description
campaign_id	UUID	PRIMARY KEY	Launch Campaign, Search Campaigns	Campaign ID
Campaign_name	VARCHAR(255)	NOT NULL	Launch Campaign	Campaign name
Campaign_Description	TEXT	NULL	Launch Campaign	Description
Campaign_start_date	DATE	NULL	Launch Campaign	Start date
Campaign_end_date	DATE	NULL	Launch Campaign	End date
Campaign_voting_date	DATE	NULL	Launch Campaign	Voting date
Campaign_status	VARCHAR(50)	NOT NULL, CHECK	Launch Campaign	Status
iCampaign_mage_url	VARCHAR(500)	NULL	Launch Campaign	Campagn image
Campaign_theme	VARCHAR(100)	NULL	Launch Campaign	Campaign Theme
campaign_goal	VARCHAR(500)	NOT NULL	Launch Campaign	Campaign Goal
Campaign_objective	VARCHAR(500)	NOT NULL	Launch Campaign	Campaign objective
created_by	VARCHAR(255)	NOT NULL	Launch Campaign	Created by user
Campaign_Success_Criteria	VARCHAR(500)	NOT NULL	Launch Campaign	Campaign Success criteria
created_at	TIMESTAMP	NOT NULL	Launch Campaign,	Created timestamp
updated_at	TIMESTAMP	NOT NULL	Launch Campaign	Updated timestamp
published_at	TIMESTAMP	NULL	Launch Campaign	Published timestamp
closed_at	TIMESTAMP	NULL	Launch Campaign	Closed timestamp

Campaign – Data Tables

Campaign_contributors

Attribute Name	Data Type	Constraints	Journey Mapping	Description
Campaign_id	UUID	PK, FK	Launch Campaign , Search Campaigns	Campaign ID
user_id	UUID	PK, FK	Launch Campaign , Search Campaigns	User ID
role	VARCHAR(50)	DEFAULT CONTRIBUTOR	Launch Campaign	Role
added_at	TIMESTAMP	DEFAULT NOW	Launch Campaign	Added timestamp
added_by	UUID	NULL	Launch Campaign	Added by

Campaign_Groups

Attribute Name	Data Type	Constraints	Journey Mapping	Description
Campaign_id	UUID	PK, FK	Launch Campaign , Search Campaigns	Campaign ID
Group_id	UUID	PK, FK	Launch Campaign , Search Campaigns	Group ID
User_id	UUID	PK, FK	Launch Campaign , Search Campaigns	User ID
added_at	TIMESTAMP	DEFAULT NOW	Launch Campaign	Added timestamp
added_by	UUID	NULL	Launch Campaign	Added by

Campaign_Distribution_List

Attribute Name	Data Type	Constraints	Journey Mapping	Description
Campaign_id	UUID	PK, FK	Launch Campaign , Search Campaigns	Campaign ID
user_id	UUID	PK, FK	Launch Campaign , Search Campaigns	User ID
Distribution_List	VARCHAR(50)	NOT NULL	Launch Campaign	Distribution List from the DB
added_at	TIMESTAMP	DEFAULT NOW	Launch Campaign	Added timestamp
added_by	UUID	NULL	Launch Campaign	Added by

Campaign_attachments

Attribute Name	Data Type	Constraints	Journey Mapping	Description
attachment_id	UUID	PRIMARY KEY	Create idea	Attachment ID
Campaign_id	UUID	NOT NULL, FK	Create idea	Idea ID
file_name	VARCHAR(255)	NOT NULL	Create idea	File name
file_url	VARCHAR(500)	NOT NULL	Create idea	File URL
file_type	VARCHAR(100)	NULL	Create idea	File type
file_size	BIGINT	NULL	Create idea	File size
uploaded_by	UUID	NOT NULL, FK	Create idea	Uploaded by
uploaded_at	TIMESTAMP	DEFAULT NOW	Create idea	Uploaded timestamp

Ideation – Data Tables

ideas

Attribute Name	Data Type	Constraints	Journey Mapping	Description
iidea_id	UUID	PRIMARY KEY	Create idea, Search Ideas	Idea ID
title	VARCHAR(255)	NOT NULL	Create idea, Search Ideas	Title
description	TEXT	NOT NULL	Create idea, Search Ideas	Description
campaign_id	UUID	NOT NULL, FK	Create idea, Search Ideas	Campaign ID
Goal_type	VARCHAR(50)	NOT NULL	Create idea	Goal Type (to be fetched from the related campaign based on idea input)
Completeness	Int	null	Create idea	This data will be generated using logic. AI suggested.(1-10)
Specificity	Int	null	Create idea	This data will be generated using logic. AI suggested.(1-10)
created_by	UUID	NOT NULL, FK	Create idea, Search Ideas	Creator
display_picture_url	VARCHAR(500)	NULL	Create idea, Search Ideas	Picture URL
status	VARCHAR(50)	NOT NULL	Create idea, Search Ideas	Status
Action	Int	null	Create idea, Search Ideas	New, Prioritize, Review, Notify, Accept, Reject, Park
Rating	DECIMAL	NULL	Create idea, Search Ideas	Rating store for Ideas
business_value	TEXT	NULL	Create idea, Search Ideas	Business value
Business_impact	VARCHAR(500)	NOT NULL	Create idea, Search Ideas	Business Impact
estimated_budget	DECIMAL	NULL	Create idea	Budget
created_at	TIMESTAMP	NOT NULL	Create idea, Search Ideas	Created timestamp
updated_at	TIMESTAMP	NOT NULL	Create idea, Search Ideas	Updated timestamp
submitted_at	TIMESTAMP	NULL	Create idea, Search Ideas	Submitted timestamp
reviewed_at	TIMESTAMP	NULL	Create idea	Reviewed timestamp
reviewed_by	UUID	NULL	Create idea	Reviewer
review_comments	TEXT	NULL	Create idea, Search Ideas	Review comments

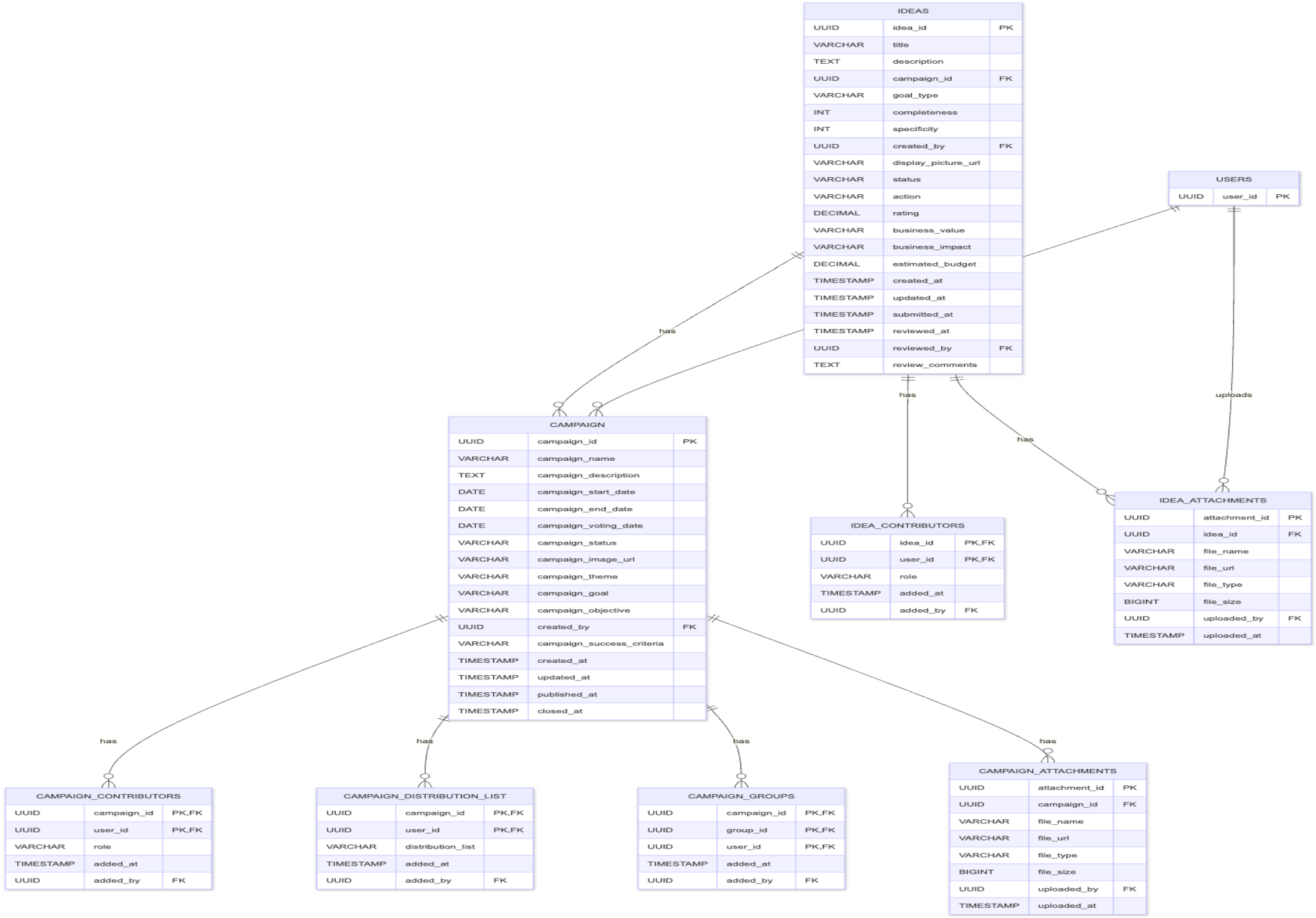
idea_contributors

Attribute Name	Data Type	Constraints	Journey Mapping	Description
idea_id	UUID	PK, FK	Create idea, Search Ideas	Idea ID
user_id	UUID	PK, FK	Create idea, Search Ideas	User ID
role	VARCHAR(50)	DEFAULT CONTRIBUTOR	Create idea	Role
added_at	TIMESTAMP	DEFAULT NOW	Create idea	Added timestamp
added_by	UUID	NULL	Create idea	Added by

idea_attachments

Attribute Name	Data Type	Constraints	Journey Mapping	Description
attachment_id	UUID	PRIMARY KEY	Create idea	Attachment ID
idea_id	UUID	NOT NULL, FK	Create idea	Idea ID
file_name	VARCHAR(255)	NOT NULL	Create idea	File name
file_url	VARCHAR(500)	NOT NULL	Create idea	File URL
file_type	VARCHAR(100)	NULL	Create idea	File type
file_size	BIGINT	NULL	Create idea	File size
uploaded_by	UUID	NOT NULL, FK	Create idea	Uploaded by
uploaded_at	TIMESTAMP	DEFAULT NOW	Create idea	Uploaded timestamp

Campaign Management – ERD



IBM